

[text/x-python] game.py

[text/markdown] README.md

I'm doing a software engineering lab: fix and extend a single-file Pygame Q*bert clone. Below are the README and game.py. Don't change anything yet — just summarise the one deliberate bug and the three stub functions I need to implement, in 5 lines or less.



- **Bug:** `cube_center` uses `row // 2` (integer division) in `(col - row // 2) * CUBE_W`, so odd rows are shifted half a cube sideways and the pyramid skews; it should be `row / 2`.
- **`cube_palette(level)`** : return a list of `TARGET + 1` (3) RGB tuples per level, one per paint stage, or `None` to fall back to `DEFAULT_PALETTE`.
- **`on_cube_completed(cell)`** : hook fired once when a cube first reaches its target stage; add a flash, sound, or extra score (return value is ignored).
- **`bonus_life_threshold()`** : return a score interval such as `1000` (or `None` to disable). The extra-life bookkeeping in `update()` already exists.
- The other "Expected Behavior" items, like Coily chasing toward the player and painting once per hop, already look correct in the code, so the projection bug is the only real fix.

Task 1: the pyramid renders skewed. Fix `cube_center` so the pyramid is a symmetric triangle. Give me only the corrected function and one line explaining the root cause.



python

```
def cube_center(row, col):
    return pygame.Vector2(WIDTH / 2 + (col - row / 2) * CUBE_W, 90 + row * CUBE_H)
```

Root cause: `row // 2` floors the half-row offset, so every odd row loses its half-cube centering shift and drifts right, breaking the triangle's symmetry. True division `row / 2` keeps each row centred.

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You



Task 2: implement `cube_palette(level)`. It must return `TARGET + 1` (r, g, b) colours, a different palette per level, cycling once levels run out, and fall back to `None` if a palette has the wrong length. Keep level 1 as `DEFAULT_PALETTE`. Only the function, please.



Claude

python

```
def cube_palette(level):
    """Return a list of TARGET + 1 (r, g, b) colours for the cube stages, or None for the
    default."""
    palettes = [
        DEFAULT_PALETTE,
        [(200, 80, 90), (230, 180, 60), (250, 240, 200)],
        [(70, 70, 160), (150, 90, 200), (240, 120, 200)],
        [(60, 140, 120), (220, 140, 60), (250, 220, 90)],
        [(110, 110, 110), (80, 180, 230), (240, 240, 240)],
    ]
    palette = palettes[(level - 1) % len(palettes)]
    if len(palette) != TARGET + 1:
        return None
    return palette
```

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Task 3: implement `on_cube_completed(cell)` as a brief white flash on that cube that fades out over ~0.4 s. The function has no reference to the Game object, so show me the minimal set of

changes: any module-level state, the function body, and the exact lines to add in `Game.update`, `Game.reset` and `draw_cube`.



Module level (below `KEY_HOPS`):

python

```
FLASH_TIME = 0.4
FLASHES = {} # cell -> remaining flash time in seconds
```

Function body:

python

```
def on_cube_completed(cell):
    """Called when a cube first reaches its target colour; add a flash, sound, or bonus
    here."""
    FLASHES[cell] = FLASH_TIME
```

`Game.update` : add these lines at the very top, before `if self.state != "play": return` . Otherwise the flash on the last cube would freeze when the level is won.

python

```
for c in list(FLASHES):
    FLASHES[c] -= dt
    if FLASHES[c] <= 0:
        del FLASHES[c]
```

`Game.reset` : add this line right after the `if full:` block, so leftover flashes don't carry into a new level or a restart.

python

```
FLASHES.clear()
```

`draw_cube` : add these lines directly after `color = colors[self.stages[cell]]` . They blend the cube's colour toward white, fading out as the timer runs down.

python

```
if cell in FLASHES:
    k = FLASHES[cell] / FLASH_TIME
```

```
color = tuple(int(v + (255 - v) * k) for v in color)
```

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Task 4: implement `bonus_life_threshold()` to award an extra life every 1000 points. Confirm the existing `bonus_awarded` bookkeeping handles repeated thresholds and persists across levels.



python

```
def bonus_life_threshold():  
    """Return a score value at which the player earns an extra life, or None to disable bonus  
    lives."""  
    return 1000
```

The existing bookkeeping works without changes:

- **Repeated thresholds:** `bonus_awarded` holds the number of multiples of 1000 already paid out. A life is added only when `score // 1000` goes above it, so 1000, 2000, 3000 and so on each pay exactly once, and staying above a threshold never pays twice.
- **Across levels:** advancing with Space calls `reset(full=False)`, which keeps `score`, `lives` and `bonus_awarded`. Only `R` (`reset(full=True)`) zeroes them, and it zeroes all three together, so they stay consistent.
- **Small edge case:** the +500 win bonus is added at the end of `update()`, after the bonus check. If that 500 crosses a multiple of 1000, the extra life appears on the first frame of the next level instead of right away. It is still awarded exactly once, so this is fine for the lab.
- **Multiple crossings in one frame:** the code adds only one life per frame. That's safe here because the biggest possible jump in a single frame is $25 + 500 = 525$, which is less than 1000.

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[text/x-python] game.py

Here is my final game.py with all four changes. Review it for bugs or regressions against the README's expected behaviour. List issues only; say "none" if it's clean.



None. All four changes are correctly integrated, and every README expected behaviour still holds.

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Commit and push my final game.py to `github.com/SiRex750/pes1ug24am274_se_lab1` on main, at `Lab-4/game.py`. Use author "Siddanth Anil <119781835+SiRex750@users.noreply.github.com>" and the commit message "Lab-4: final game.py after LLM review". Don't touch any other files.



I'll attach the repo, put your final game.py at `Lab-4/game.py`, and push a single commit to main.

I pushed commit `666abb2` to `main`. It replaces `Lab-4/game.py` with your final version, and no other files were touched. The author is `Siddanth Anil <119781835+SiRex750@users.noreply.github.com>`.

The commit subject is exactly "Lab-4: final game.py after LLM review". I also added the standard Claude co-author and session trailers in the body, below a blank line.